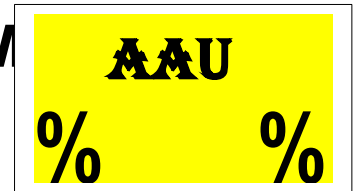


DEPARTMENT OF ANATOMY

FACULTY OF MEDICINE ADDIS ABABA UNIVERSITY



HISTOLOGY & EMBRYOLOGY FINAL EXAMINATION FOR YEAR I MEDICAL STUDENTS

Name: _____
Date: January 17, 2011

Time allowed: 2:30 hrs
Total No of pages= 12

Histology - 60%

For each of the following Multiple Choice Questions, choose the one best appropriate answer and mark "X" in the appropriate box located against the corresponding question number and letter in the separate answer sheet provided. *(60 points, 1 point for each correct answer)*

1. With regard to the cellular structures, which statement is **false**?
a) Peroxisomes contain oxidative enzymes. b) Microtubules are more stable than microfilaments.
c) Glycolipids are present in the outer leaflet of plasmalemma.
d) Golgi apparatus sorts and packages proteins within a cell.
e) Centrioles self-duplicate during the S phase of a cycle.
2. Where are ribosomes constructed?
a. Cytoskeleton b. Peroxisome c. Microfilaments d. Centrioles e. Nucleoli
3. Which of the following is NOT a membranous organelle?
a. Lysosomes b. Peroxisomes c. Centrioles d. Mitochondria e. Endoplasmic reticulum
4. Which of the following is the longest and most variable cell-cycle phase in continuously dividing cells?
a) G₀ b) G₁ c) G₂ d) M e) S
5. Concerning the connective tissue, which statement is correct?
a. Dura mater is composed of dense irregular connective tissue.
b. Primary bone has a higher mineral content than the secondary bone.
c. Elastic cartilage lacks an identifiable perichondrium.
d. Brown adipose tissue tends to increase as humans age.
2. Dense regular connective tissue contains more mast cells than any other type of connective tissue types.
6. In the connective tissue, which statement is not correct?
a. Costal cartilage are made up of hyaline cartilage. b. Osteon is unmineralized bone matrix.
c. The collagen in elastic cartilage matrix is mainly of type II. d. Osteoclast is involved in bone resorption.
e. Sharpey's fibers are collagen fibers that extend into a bone at an angle.
7. Which **type of cartilage** forms the malleus (hammer), incus (anvil) and stapes (stirrup) of the middle ear?
a. Hyaline cartilage b. Elastic cartilage c. Fibrocartilage d. All of them e. None of them

8. Pertaining to the **connective tissue cells**,

- a) pericytes originate in the bone marrow as monocytes.
- b) mast cells arise from myeloid stem cells during hematopoiesis.
- c) adipose cells are responsible for the synthesis and storage of fat.
- d) fibroblasts are located mostly along the capillaries.
- e) macrophages moderate allergic reactions.

9. In the **neurons**,

- a) dendrites may contain Nissl bodies, but lack Golgi complex.
- b) neurotransmitters are produced, stored and released by Schwann cells.
- c) nodes of Ranvier are encapsulated aggregations of neuronal cell bodies in the CNS.
- d) the nucleus contains multiple nucleoli.
- b) the axon hillock contains Nissl bodies.

10. With regard to the **neuroglial cells**, which statement is **true** ?

- a) All neuroglial cells can generate as many as 1000 action potentials per second.
- b) Oligodendrocytes are located in both gray matter and white matter.
- c) Microglia is the central nervous system macrophage.
- d) Astrocytes produce myelin.
- e) Ependymal cells help to establish the blood-brain barrier.

11. Which of the following cell types may provide myelin for several axons simultaneously?

- a) ependymal cells
- b) oligodendrocytes
- c) microglia
- d) Schwann cells
- e) satellite cells

12. Select the **correct** statement about the **nerve tissue**.

- a) Sensory neurons are unipolar.
- b) Fibrous astrocytes are found predominantly in white matter.
- c) Satellite cells surround neuronal perikaryons located in the ganglia.
- d) All of them
- e) None of them

13. In the **urinary system**, which statement is **true**?

- a) The space between the renal pyramids is called nephron.
- b) The phagocytic juxtaglomerular cells are present in the interstitium of the renal glomeruli.
- c) Transitional epithelium composes the kidney tubules.
- d) The ascending thick limb of the loop of Henle is impermeable to water even in the presence of ADH.
- e) Macula densa cells are sensitive to sodium concentration.

14. Concerning the **urinary system**, which statement is **true**?

- a) the distal convoluted tubules possess brush borders and apical canaliculi.
- b) macula densa are located in the urinary pole of the nephrons.
- c) podocytes are highly modified neurons that form the parietal layer of Bowman's capsule.
- d) papillary collecting tubules empty at area cribosa.
- e) the female ureter is lined primarily by stratified squamous epithelium.

15. In the **urinary system**, which statement is **false**?

- a) A nephron and its collecting tubules constitute the seminiferous tubule.
- b) Extraglomerular mesangial cells are tall, narrow, closely packed epithelial cells in the distal tubule.
- c) The major and minor calyces are lined by transitional epithelium.
- d) Antidiuretic hormone makes the epithelium of collecting duct permeable to water.
- e) Glomerular capillaries are lined by fenestrated endothelium.

16. With regard to the **urinary system**, which statement is **true**?

- a). Vasa recta are seen in the renal cortex.
- b) Juxtaglomerular apparatus is located at the urinary pole of the renal corpuscle.
- c) The renal cortical tissue that extends between the pyramids is called renal column of Bertin.
- d) Distal convoluted tubule is longer and more convoluted than the proximal convoluted tubule.
- e) The diameter of afferent arteriole of the glomerulus is smaller than that of efferent arteriole.

17. Which of the following hormones is carried to its target(s) by the hypophyseal portal system?
 a) ADH b) ACTH c) FSH d) TRH e) TSH
18. With regard to the **endocrine system**, which statement is **true**?
 a. The adrenal cortex is essential to life.
 b. Hormones must be present at high concentrations to elicit an effect.
 c. Oxyphil cells are small basophilic cells arranged in clusters within the stroma of the parathyroid gland that secrete calcitonin.
 d. Pituicytes are undifferentiated cells that are capable of differentiating into various types of chromophils.
 e. Releasing and inhibiting hormones are stored in Herring bodies.
19. Which of the following statements is **true** in the **endocrine system**?
 a. Testosterone is a protein hormone. b. Somatotrophic cells secrete ACTH.
 c. Pineal gland secretes melatonin. d. Zona glomerulosa of the adrenal gland secretes glucocorticoids.
 e. Parafollicular cells are involved in the secretion of thyroglobulin.
20. With regard to the **endocrine system**, which statement is **false**?
 a) Lactotropic cells can be classified as acidophil.
 b) Chromaffin cells of the adrenal medulla synthesize, store and secrete catecholamines.
 c) The pars tuberalis of adenohypophysis is composed of cuboidal basophilic cells.
 d) The pineal gland has a dense irregular collagenous connective tissue capsule formed from dura mater.
 e) Vasopressin (ADH) acts as a powerful vasoconstrictor of smooth muscle cells in the tunica media of arterioles.
21. In the **male reproductive system**, which statement is **false**?
 a) There are two corpora cavernosa and one corpus spongiosum in the penis.
 b) The prostate is made of compound tubuloalveolar glands.
 c) the ductus epididymidis is lined by pseudostratified columnar epithelium.
 d) The hormone produced by interstitial cells of Leydig is testosterone.
 e) during spermiogenesis, the spermatids quickly undergo the second meiotic division to give spermatozoa.
22. What gland in the male is homologous to the greater vestibular gland in the female?
 a. Bartholin's gland b. Prostate gland c. Skene's gland d. Cowper's gland e. Brunner's gland
23. In the **male reproductive system**, which statement is **true**?
 a) Sertoli cells possess receptors for LH.
 b) Ductus epididymidis secretes glycerophosphocholine.
 c) Ejaculatory duct has a very thick muscular wall.
 d) Seminal vesicles contain corpora amylacea.
 e) Interstitial cells of Leydig phagocytose excess cytoplasm discarded by maturing spermatids.
24. In the **female reproductive system**, which statement is **false**?
 a) secondary follicles are dependent on FSH for their continued development.
 b) Tunica albuginea is the connective tissue which surrounds the ovary.
 c) Fornix of vagina is lined by stratified squamous epithelium.
 d) The vagina lacks glands throughout its length and is lubricated by secretions from cervical glands.
 e) Germinal epithelium or the surface epithelium covering the ovary gives rise to ova.
25. During a pregnancy, which hormone maintains the **corpus luteum**?
 a. FSH b. LH c. hCG d. Estrogen e. Progesterone

26. In the **female reproductive system**, select the **incorrect** statement.

- a) Primary oocyte completes its first meiotic division at the time of ovulation.
- b) The basal layer of the endometrium is the thick smooth muscle tunic of the uterus.
- c) Uterine tube is lined by simple columnar ciliated epithelium.
- d) Mammary glands are compound tubuloalveolar glands.
- e) Theca interna cells in the ovaries manufacture androgens.

27. In the **eye**, which statement is **false**?

- a) The ciliary body generates the aqueous humor.
- b) Ora serrata is the anterior pigmented portion of the retina.
- c) Bruch's membrane is located between the choroid and the retina.
- d) The cornea is covered by an incomplete layer of pigmented cells and fibroblasts on its anterior surface.
- e) Lacrimal canaliculi are lined by a stratified squamous epithelium.

28. Which structure of the **eye** is avascular?

- a. Retina
- b. Cornea
- c. Choroid
- d. Sclera
- e. Uvea

29. With regard to the **ear**, which statement is FALSE?

- a) the bony labyrinth is filled with endolymph.
- b) sensory neuroepithelial regions of the semicircular canals from the cristae ampullaris.
- c) the external auditory meatus is lined by keratinized stratified squamous epithelium.
- d) pinna consists of an irregular piece of elastic cartilage surrounded by a thick perichondrium.
- e) the lining epithelium of the tympanic membrane is ciliated columnar interspersed with ceruminous glands.

30. With regard to the **ear**, one of the following statements is **false**.

- a) The middle ear contains ossicles.
- b) Stria vascularis secretes endolymph.
- c) The bony labyrinth is filled with endolymph.
- d) The vestibule houses the saccule and utricle.
- e) Endolymphatic sac contains phagocytic cells like macrophages and neutrophils.

EMBRYOLOGY - 40%

For each of the following **Multiple Choice Questions** choose the **one best appropriate answer** and **mark "X"** in the appropriate box located against the corresponding question number and letter in the **separate answer sheet** provided. *(40 points, 1 point for each correct answer)*

1. With regard to **gametogenesis**, which statement is **false**?
 - a) All primary oocytes are formed prenatally.
 - b) Primary oocytes remain dormant in prophase of meiosis I until puberty.
 - c) If fertilization does not occur, the secondary oocyte becomes primary oocyte.
 - d) The primary spermatocytes are the largest germ cells in the seminiferous tubules.
 - e) The difference in the sex chromosome complement of sperms forms the basis of primary sex determination.
2. Which statement is **true** about **female reproductive cycles**?
 - a) LH stimulates the development of ovarian follicles and the production of estrogen by its follicular cells.
 - b) ovulation is triggered by a surge of FSH production.
 - c) if an oocyte is fertilized, the corpus luteum involutes and degenerates.
 - d) during the proliferative phase, the corpus luteum attains the maximum growth.
 - e) ischemic phase occurs when the oocyte is fertilized.
3. Which statement is **false** about **fertilization**?
 - a) it restores the normal diploid number of chromosomes.
 - b) it stimulates the penetrated oocyte to complete the first meiotic division.
 - c) it causes metabolic activation of the ootid and initiates cleavage of the zygote.
 - d) it results in variation of the human species.
 - e) it determines the chromosomal sex of the embryo.
4. Where does the **blastocyst normally implant**?
 - a) functional layer of the cervix
 - b) myometrium
 - c) functional layer of the endometrium
 - d) perimetrium
 - e) basal layer of the endometrium
5. Which of the following structures must degenerate for **blastocyst implantation** to occur?
 - a) endometrium in progestational phase
 - b) zona pellucida
 - c) syncytiotrophoblast
 - d) cytotrophoblast
 - e) functional layer of the endometrium.
6. Select the **incorrect** statement.
 - a) gastrulation is first indicated by the formation of the primitive streak.
 - b) neurulation is the process by which neuroectoderm forms the neural plate.
 - c) preponderance of sacrococcygeal teratoma is found in male newborns.
 - d) epiblast and hypoblast are fused in the cloacal membrane.
 - e) all germ layers are derived from epiblast.
7. Which of the following is/are **not** a derivative(s) of the **neural crest**?
 - a) Schwann cells
 - b) chromaffin cells
 - c) neurohypophysis
 - d) melanocytes
 - e) parafollicular (C) cells.
8. Which structure is/are **not** visible during the **fourth week**?
 - a) closed neuropores
 - b) three pairs of pharyngeal arches
 - c) scalp vascular plexus
 - d) upper limb buds
 - e) lens placodes.

9. Select the **correct** event during the **fetal period**.

- a) fetal spleen is an important site of hematopoiesis during week 20.
- b) the quantity of white fat is about 8% of body weight during week 21.
- c) during the 10th week, the skin of the fetus is covered by vernix caseosa.
- d) the eyelids open at week 18.
- e) umbilical herniation is a normal event during the 38th week.

10. Which of the following is not **false** about **birth defects**?

- a. Short stature in females is one of the classic features of the Turner syndrome.
- b. Tetracycline would cause stained teeth and hypoplasia of enamel.
- c. A weak catlike cry and microcephaly indicate the clinical diagnosis of achondroplasia.
- d. Sparse body hair and gynecomastia may indicate the clinical diagnosis of Klinefelter's syndrome.
- e. Thalidomide induces limb defects.

11. With regard to the **development of the skull**, which of the following statements is **false**?

- a. The membranous portion of the skull is derived from neural crest cells and paraxial mesoderm.
- b. The parachordal cartilage contributes to the base of the occipital bone and forms the boundaries of the foramen magnum.
- c. The hypophyseal cartilage forms the petrous and mastoid parts of the temporal bone.
- d. The trabeculae cranii fuse to form the body of the ethmoid bone
- e. Nasal capsules contribute to the formation of the ethmoid bone.

12. Which of the following statements is **true** about the **development of muscles**?

- a. The intercostal muscles remain segmentally arranged like the somites.
- b. The extrinsic eye muscles are innervated by the hypoglossal nerve (CN XII).
- c. The first pair occipital (postotic) myotomes form the muscles of mastication.
- d. Myoblasts of the hypomeres form the extensor muscles of the vertebral column.
- e. The sacrococcygeal myotomes of epaxial divisions form the muscles of the pelvic diaphragm and the striated muscles of the anus and sex organs.

13. Concerning the development of the **urinary system**,

- a) Nephrons are derived from ureteric buds.
- b) The kidneys come in contact with the suprarenal glands at birth, and the “ascent” stops.
- c) Bilateral renal agenesis is associated with oligohydramnios.
- d) The urinary bladder enters the pelvis minor during the first trimester.
- e) The podocytes of Bowman’s capsule develop from the metanephric vesicles.

14. The **proximal convoluted tubules** of the definitive adult kidney are derived from

- a) ureteric bud
- b) metanephric vesicle
- c) metanephric duct
- d) pronephric tubules
- e) metanephric tubules

15. The **labia minora** arise embryologically from

- a) phallus
- b) labioscrotal swellings
- c) sinovaginal bulbs
- d) urogenital folds
- e) paramesonephric duct

16. With regard to the development of the **genital system**, which statement is **false**?

- a) Seminal colliculus is the adult derivative of the sinus tubercle.
- b) Sertoli cells constitute most of the seminiferous epithelium in the fetal testis.
- c) Testosterone stimulates the development of the mesonephric ducts (vas deferens - epididymis).
- d) Müllerian inhibiting substance (MIS) suppresses the para - mesonephric ducts (female duct system).
- e) None of the above.

17. Concerning the development of the **pharyngeal apparatus**, which statement is **false**?

- a) Ventral end of the 2nd arch cartilage ossifies to form the stapes of the middle ear and the styloid process of the temporal bone.
- b) Musculature of the 1st pharyngeal arch forms the muscles of mastication.
- c) The inferior parathyroid gland is derived from the 3rd pharyngeal pouch.
- d) Glossopharyngeal nerve is the 3rd pharyngeal arch cranial nerve.
- e) Pharyngeal groove 1 gives rise to the external acoustic meatus.

18. With regard to the development of the **endocrine system**,

- a) The thyroid gland develops from the second pharyngeal pouch.
- b) The pineal gland is the first endocrine gland to develop during the embryonic period.
- c) Only the zona glomerulosa and the zona fasciculata are present in a newborn infants.
- d) The adenohypophysis develops from mesoderm of diencephalon and the neurohypophysis is formed from the neural crest cells.
- e) The oxyphil cells of parathyroid gland differentiate during the embryonic period and become functionally active in regulating fetal calcium metabolism.

19. Which **cranial nerve** does **not innervate the tongue**?

- a) Trigeminal nerve
- b) Facial nerve
- c) Glossopharyngeal nerve
- d) Vestibulocochlear nerve
- e) Vagus nerve

20. The intermaxillary segment does not give rise to:

- a) Vertical groove of the upper lip (philtrum)
- b) Premaxillary part of the maxilla.
- c) Gingiva.
- d) Primary palate.
- e) Auricular hillocks.